EE 656 / EE 756 Big Data Analytics

Assignment #6: Using Hive for Data Analytics
Due Date: 8/8/2025

Overview

This assignment focuses on using Apache Hive to analyze the MovieLens dataset using the u.data, u.item, and u.user files. You will create and query Hive tables, practice joins, use built-in functions, and perform analytical queries.

Dataset Files (from the MovieLens dataset)

- u.data – (user_id, movie_id, rating, timestamp)
- u.item – (movie_id, title, release_date, ..., 24 fields total)
- u.user – (user_id, age, gender, occupation, zip_code)

Part 1: ~~Given the SQL statement, Write it in MapReduce~~ Create the table in HIVE

a) Create the ratings table and load data from HDFS
b) Create the movies table and load data from HDFS
c) Create the users table and load data from HDFS

Part 2: Basic Queries and Joins

a) Show the first 10 records from each table
b) Retrieve all ratings made by users under the age of 25 (Include movie title and rating value)
c) List all movies rated 5 by female users (Join ratings, users, and movies)
d) Find the average rating for each movie (Return movie title and average rating)
e) Count how many movies each user has rated (Return user_id and count)

Part 3: Aggregation and Filtering

a) Top 10 movies by average rating (min. 100 ratings). Return movie title, number of ratings, and average rating.
b) Most active users. List the top 5 users who have given the most ratings.
c) Average rating by occupation. Return occupation and average rating.
d) Distribution of ratings (1–5). Count how many times each rating value appears.

Part 4: Hive Features and Analytics

a) Create a view: highly_rated_movies. Include movies with average rating > 4.5 and at least 50 ratings.

Turn Over: Review Video + SQL File